

Calibrating Uncertainty for Zero-Shot Adversarial CLIP

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AIP

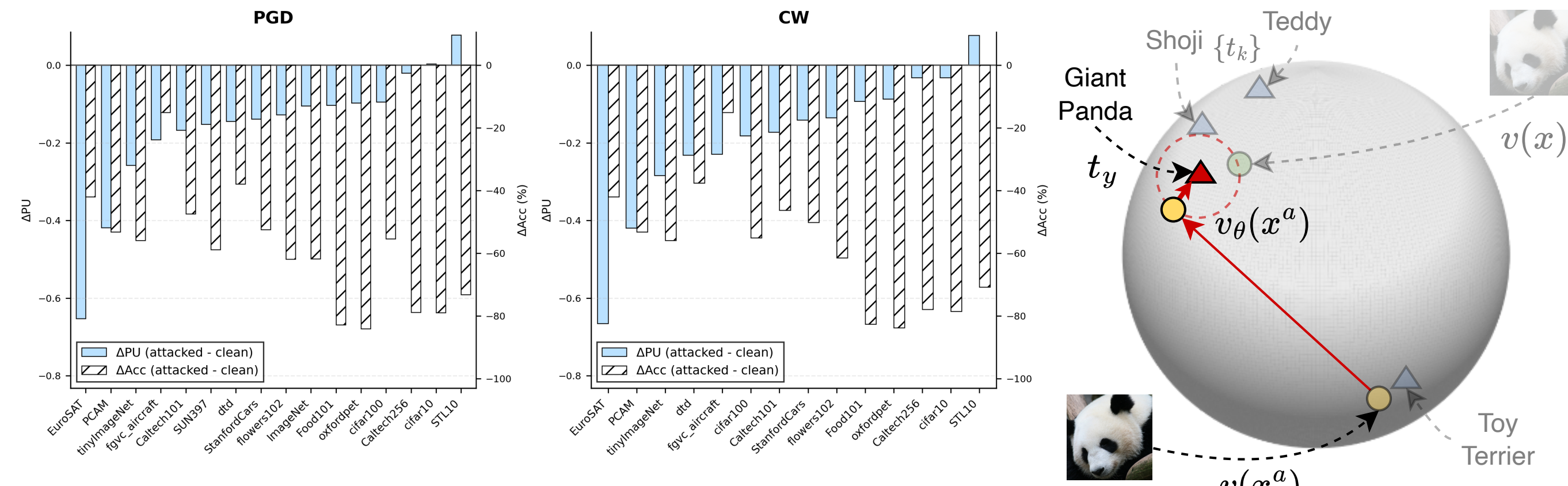


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When CLIP Is Misled, *It Should Be Uncertain*.

Motivation: When CLIP is misled, it becomes too certain.

Adversarial CLIP can become **less accurate but more certain**, causing **over-confident** errors and severe **miscalibration**.



Single-anchor alignment.

1. Perturbations induce **semantic shifts** on CLIP's shared hypersphere.
2. Existing **ZSAR** relies on **single-anchor** alignment, ignoring **class relations** and **evidence strength**.

Preliminary: From CLIP Similarity to Predictive Evidence

Softmax tells **which class wins**.

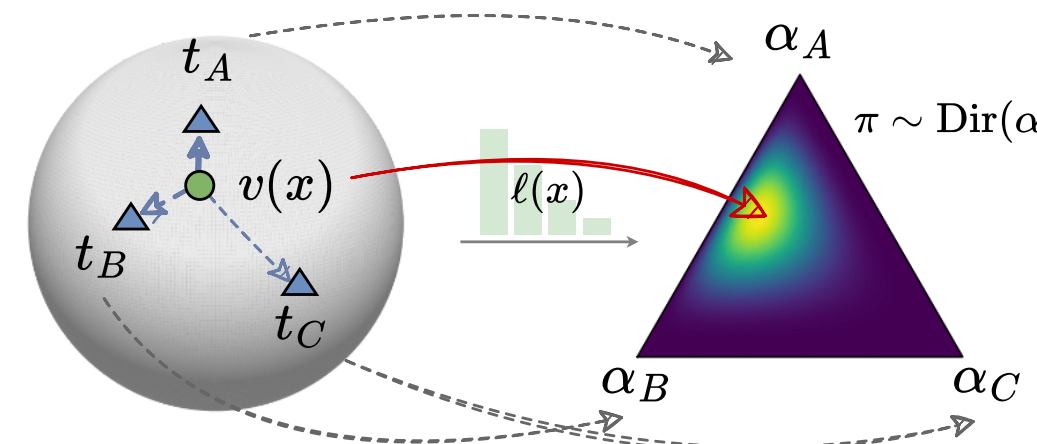
A Dirichlet lift **also** tells **how much evidence supports** the prediction.

CLIP Geometry.

$$z_k(x) = \langle v(x), t_k \rangle \in [-1, 1], \quad \ell_k(x) = z_k(x)/\tau.$$

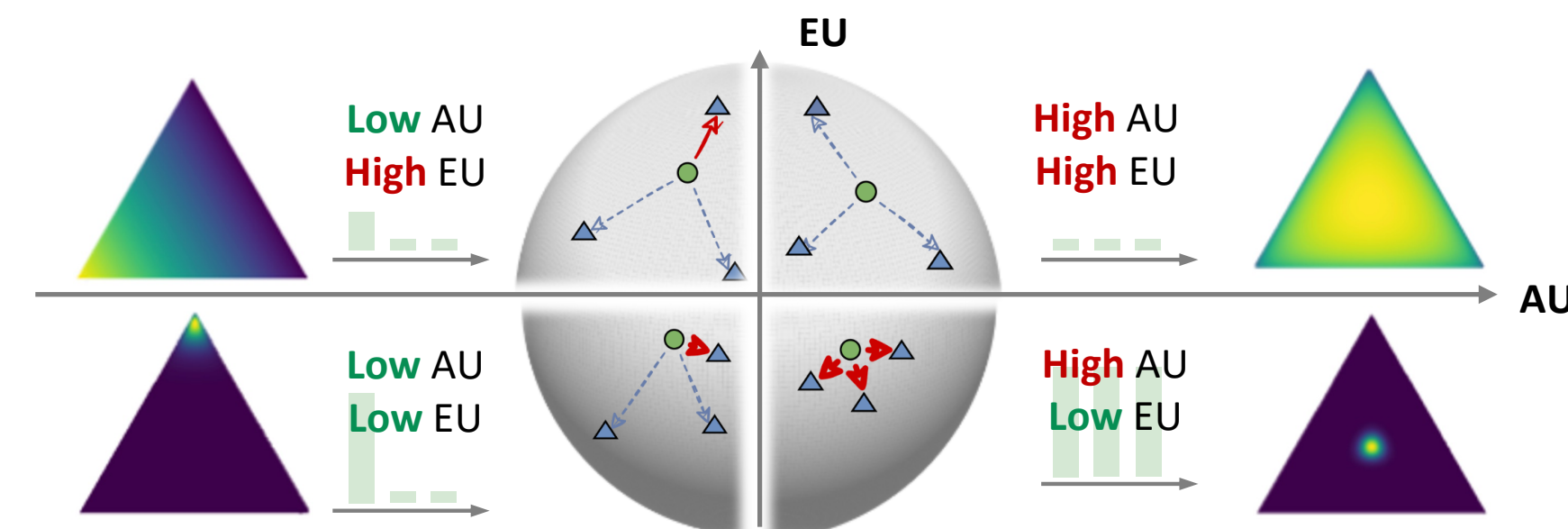
Dirichlet Evidence.

$$\alpha_k = \exp(h(\ell_k)), \quad \pi \sim \text{Dir}(\alpha), \quad \alpha_0 = \sum_j \alpha_j.$$



Uncertainty Decomposition.

Representation Geometry $\rightarrow$ Dirichlet Distribution.



AU Measures Ambiguity. EU Measures Trust.

Aleatoric uncertainty (AU)

$$\text{AU} = -\sum_k \frac{\alpha_k}{\alpha_0} [\psi(\alpha_k + 1) - \psi(\alpha_0 + 1)].$$

Epistemic uncertainty (EU)

$$\text{EU} = \frac{C}{\alpha_0 + C}.$$

Core Theory: A Faithful Dirichlet Reformulation of CLIP

The reformulation is exact:
It adds **uncertainty structure** without changing CLIP's zero-shot predictor.

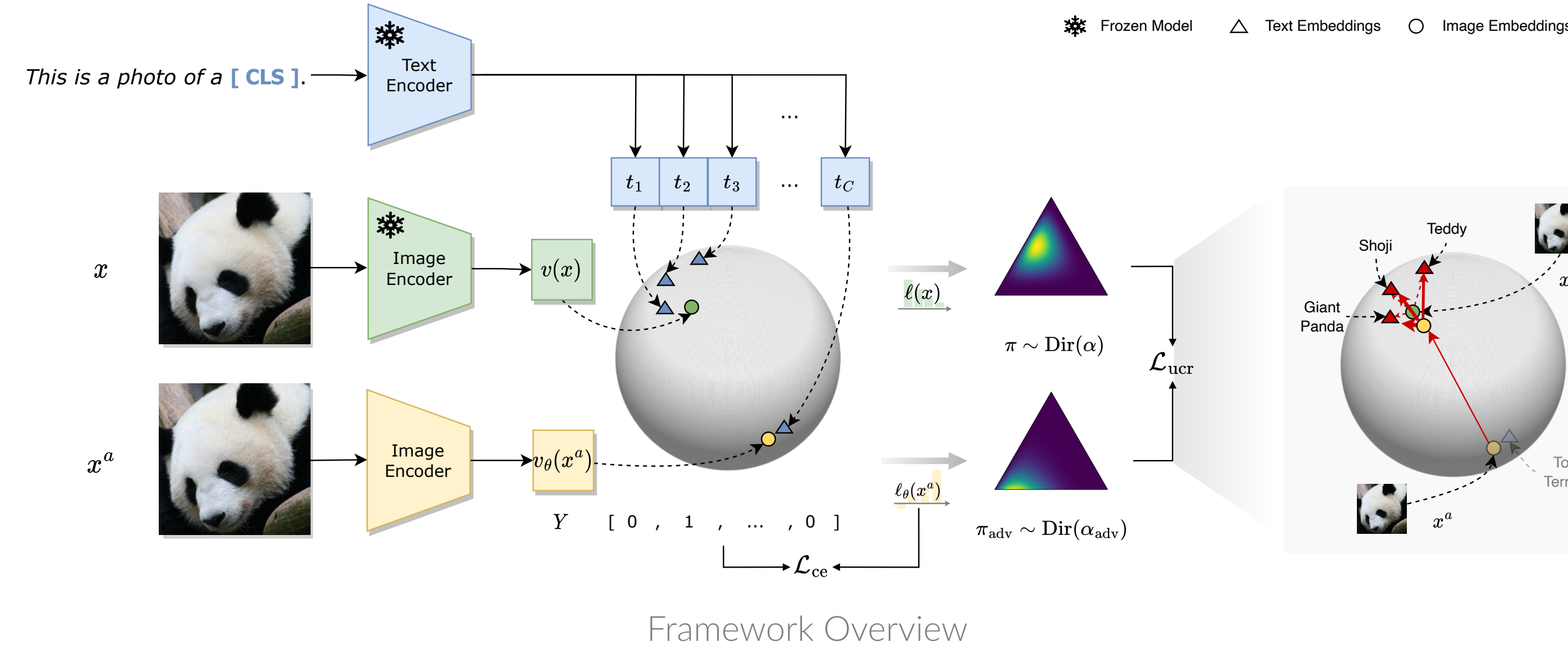
$$\underbrace{\langle v(x), t_k \rangle}_{\text{cosine similarity}} \xrightarrow[\text{shift (+1), scale (1/\tau')}]{} \underbrace{\alpha_k(x)}_{\text{Dirichlet Concentration}} = \exp\left(\frac{\tau \ell_k(x) + 1}{\tau'}\right)$$

$v(x)$: image embedding, t_k : class text embedding, $\ell_k(x) = \langle v(x), t_k \rangle / \tau$,
 τ : CLIP scale, τ' : calibration temperature.

- **Key idea:** transform CLIP similarities into **positive, monotone** Dirichlet concentration with **tunable calibration**.

Methodology: Uncertainty-Calibrated Adversarial fine-Tuning (UCAT)

Fine-tune the model to preserve **global semantic relation** and **absolute evidence strength** under adversarial perturbations.



1. Adversarial Discrimination

$$\mathcal{L}_{\text{ce}} = -\log \frac{\exp(\langle v(x^a), t_y \rangle / \tau)}{\sum_{j=1}^C \exp(\langle v(x^a), t_j \rangle / \tau)}.$$

Keeps adv. images aligned with target texts.

2. Distributional Calibration

$$\mathcal{L}_{\text{ucr}} = \text{KL}[\text{Dir}(\alpha_{\text{adv}}) \parallel \text{Dir}(\alpha)]. \quad (\text{Reverse KL})$$

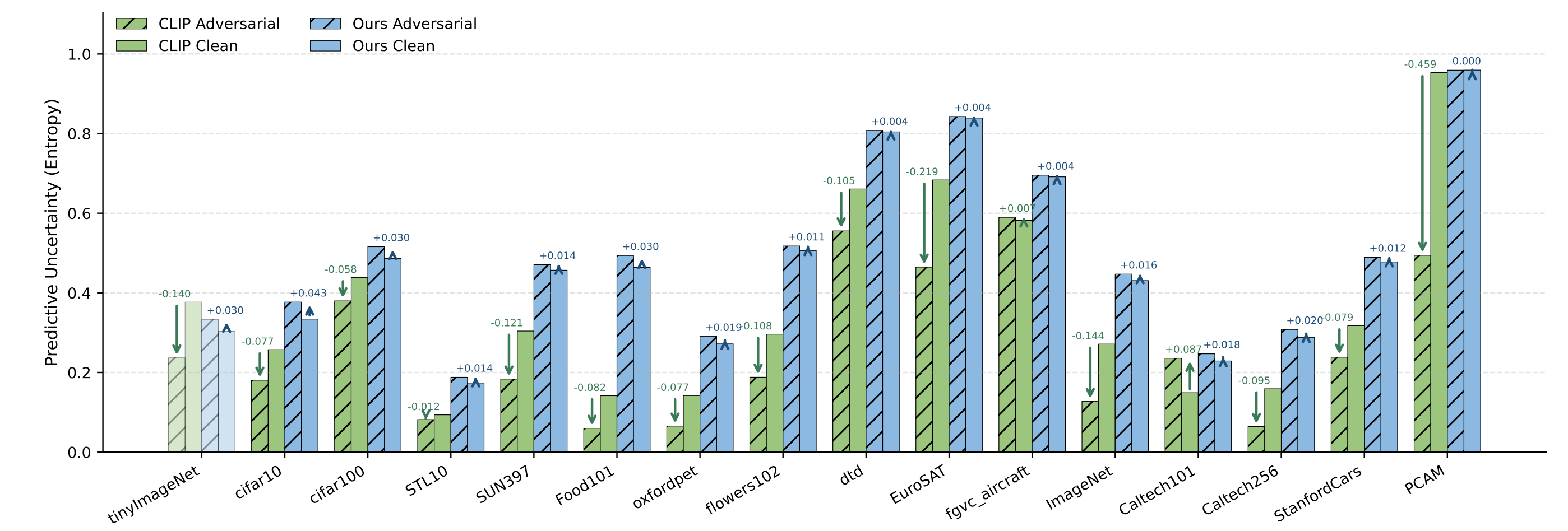
Aligns **relative semantics** (AU) and **absolute evidence** (EU) to suppress confident errors.

$$\mathcal{L}_{\text{UCAT}} = \underbrace{\mathcal{L}_{\text{ce}}}_{\text{Adversarial Discrimination}} + \lambda \underbrace{\mathcal{L}_{\text{ucr}}}_{\text{Dirichlet Calibration}}$$

Experiments: Better Robustness & Calibration

UCAT restores the desired risk order:
adversarial examples **become more uncertain** than clean ones.

Orig. w/ Clean < **Fine-tuned w/ Clean** < **Fine-tuned w/ Adv.**



Predictive uncertainty under clean/adversarial evaluation on 16 datasets.

Table 1. Single-label robustness.

Average over 16 zero-shot datasets.

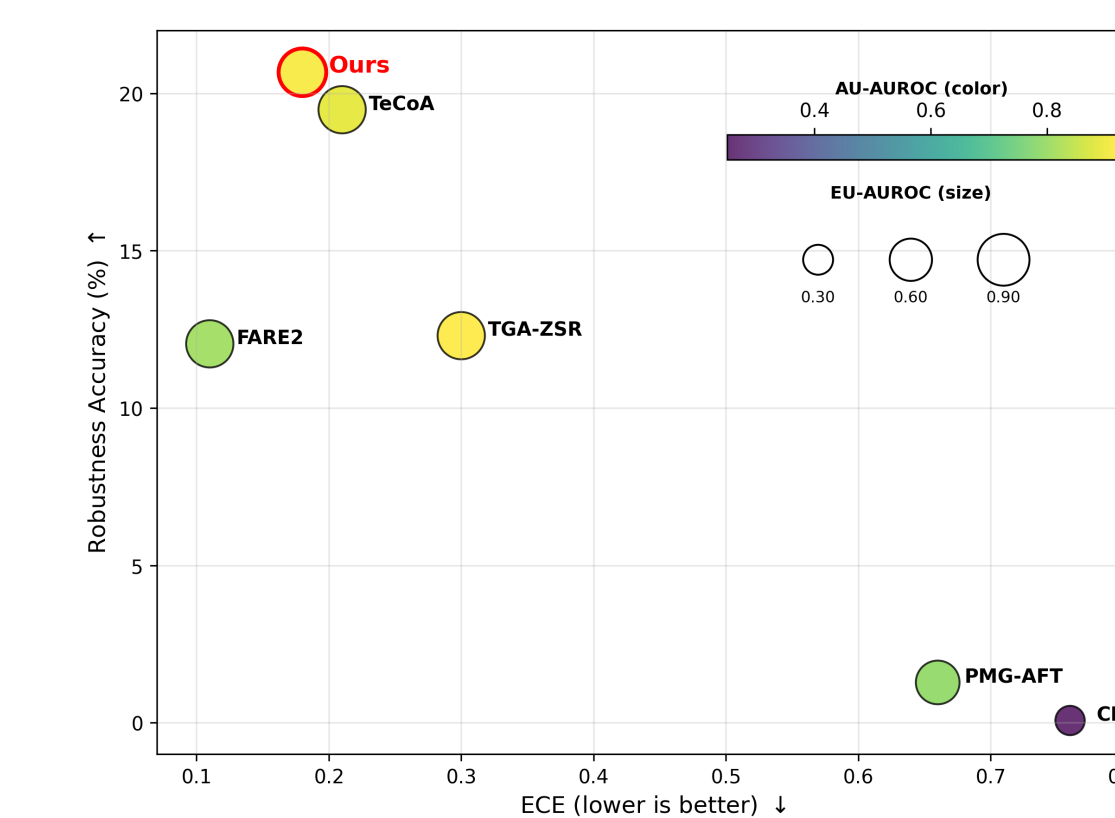
Method	Clean	Robust Accuracy					
		PGD	CW	AA	CAA	A ³	
CLIP	64.45	0.00	0.27	0.07	5.32	5.58	
TeCoA	26.72	20.73	19.96	19.48	20.59	20.52	
FARE	21.41	13.14	12.13	12.04	12.99	12.95	
PMG-AFT	53.32	11.86	4.47	2.06	11.54	11.83	
TGA-ZSR	53.25	20.02	19.86	12.30	14.72	19.64	
Comp-TGA	51.48	18.84	18.21	11.87	12.48	18.44	
UCAT	45.21	22.45	21.50	20.67	22.21	22.09	

Table 2. Multi-label robustness. MS-COCO

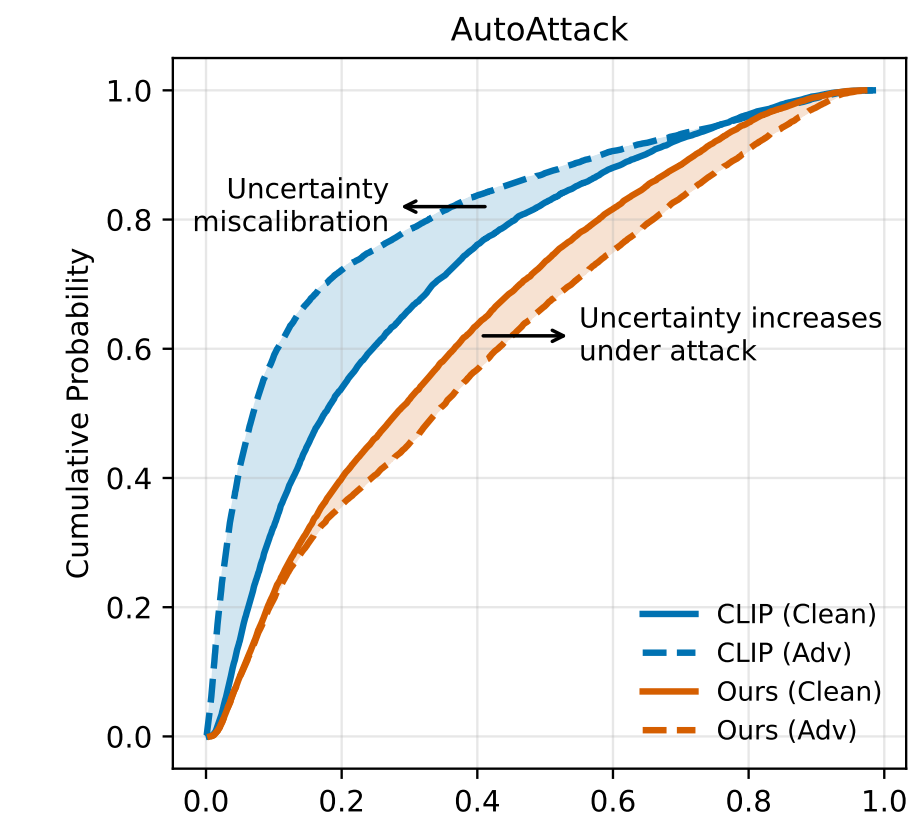
F1@3 under PGD and CW.

Method	Clean	$\epsilon = 1/255$		$\epsilon = 2/255$		$\epsilon = 4/255$	
		PGD	CW	PGD	CW	PGD	CW
CLIP	45.90	40.93	25.52	28.19	10.19	15.74	3.89
TeCoA	34.14	39.63	30.60	30.92	25.16	25.87	19.91
FARE	44.37	39.11	33.86	33.30	22.72	24.61	13.38
PMG-AFT	42.29	36.44	32.72	33.32	23.67	25.47	13.62
TGA-ZSR	37.65	32.73	33.36	33.93	28.04	27.65	18.87
Comp-TGA	38.58	36.54	33.45	33.73	28.51	27.30	18.30
UCAT	42.07	40.60	37.04	37.94	31.78	30.32	21.59

- Improves **zero-shot adv. robustness** with competitive clean accuracy.
- Achieves the best ECE-accuracy trade-off.



Robustness vs. Calibration (ECE).



PU CDFs under AA.

